

Sequence-pair representation

Sequence pair uses positive permutation A B C E D and negative D A B C E

Pseudocode

- Sequence-pair pseudocode turns two permutations into horizontal and vertical constraints
- Open this module's examples file and find the Pseudocode section
- That written sketch is what you implement on the implement track and what the browser

Algorithm sketch

- Golden pos is A B C E D and neg is D A B C E
- Evaluation must place all five modules legally with non-negative coordinates

Algorithm sketch — try these

```
INPUT: pos[], neg[] permutations of modules  
OUTPUT: (x,y) via constraint longest paths  
pos/neg order  $\Rightarrow$  horizontal & vertical constraints  
 $x \leftarrow$  longest path in H-graph;  $y \leftarrow$  V-graph  
GOLDEN pos=A B C E D; neg=D A B C E  
packs legally with non-negative coords
```

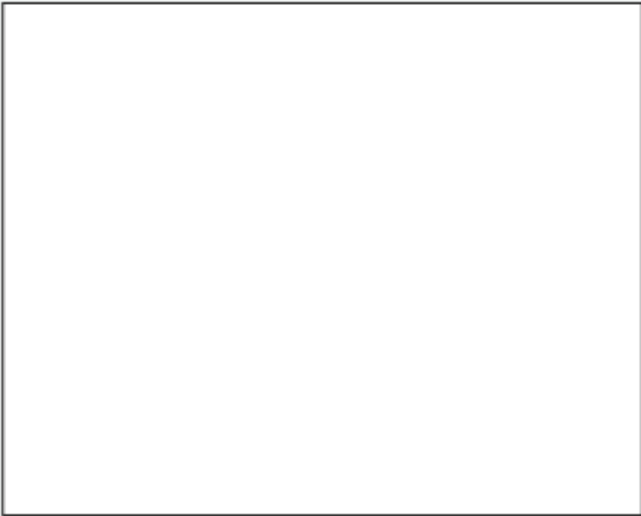
Two permutations encode geometry

SEQUENCE PAIR

Two permutations encode geometry

Step 1 / 5 · seqs · module02-05-sequence-pair

10×8



Explanation

Sequence pair uses a positive and a negative permutation of the same modules. Ours are pos A B C E D and neg D A B C E—five ids each, same set.

- pos + neg are permutations
- Same five module ids
- Constraints → longest-path pack

pos: A B C E D

neg: D A B C E

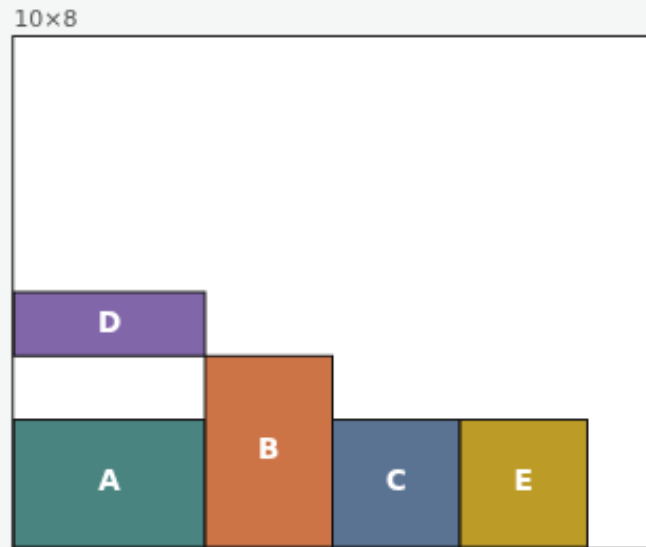
Two permutations encode geometry

Horizontal constraints set x

SEQUENCE PAIR

Horizontal constraints set x

Step 2 / 5 · h-rules · module02-05-sequence-pair



Explanation

Module i is left of j when i appears before j in both sequences. Longest-path packing pushes each block as far left as those constraints allow.

- i left of j if $\text{pos}(i) < \text{pos}(j)$ and $\text{neg}(i) < \text{neg}(j)$
- x = longest path over left-of edges

A tends left
E tends right

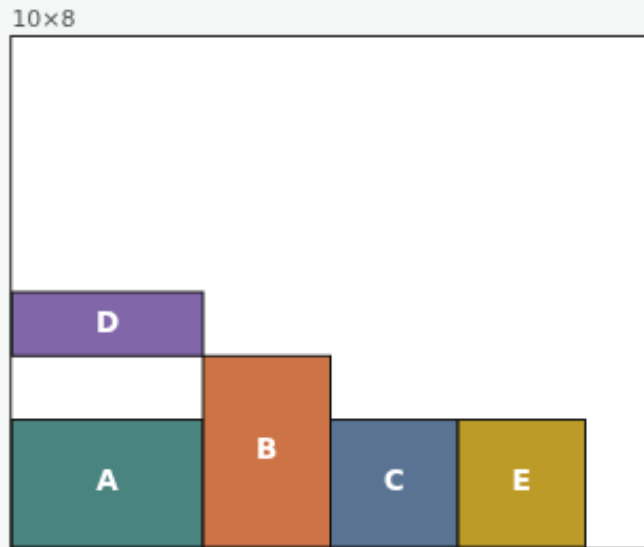
Horizontal constraints set x

Vertical constraints set y

SEQUENCE PAIR

Vertical constraints set y

Step 3 / 5 · v-rules · module02-05-sequence-pair



Explanation

Module i is below j when i precedes j in pos but follows in neg. That fills y coordinates without overlaps for a feasible pair.

- i below j if $\text{pos}(i) < \text{pos}(j)$ and $\text{neg}(i) > \text{neg}(j)$
- y = longest path over below edges

D.y=3

D can sit above the bottom row

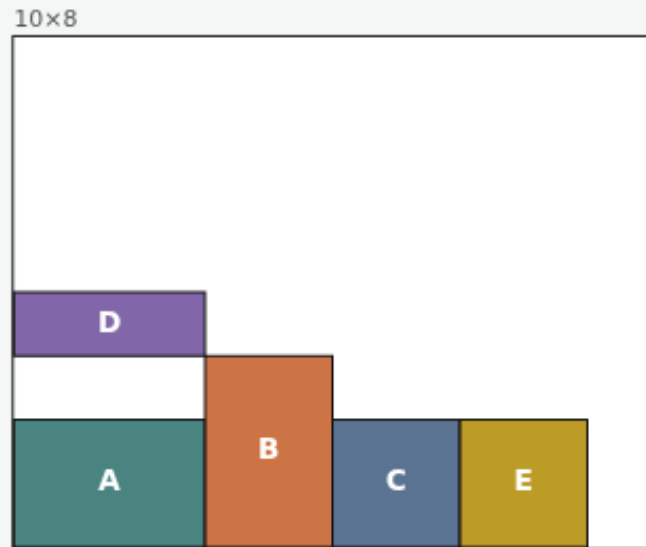
Vertical constraints set y

Golden SP packs legally

SEQUENCE PAIR

Golden SP packs legally

Step 4 / 5 · legal · module02-05-sequence-pair



Explanation

Evaluating the golden sequences places all five modules with non-negative coordinates and a legal packing inside the outline.

- Five modules placed
- legal: true
- Coords ≥ 0

legal: true
ids: A-E

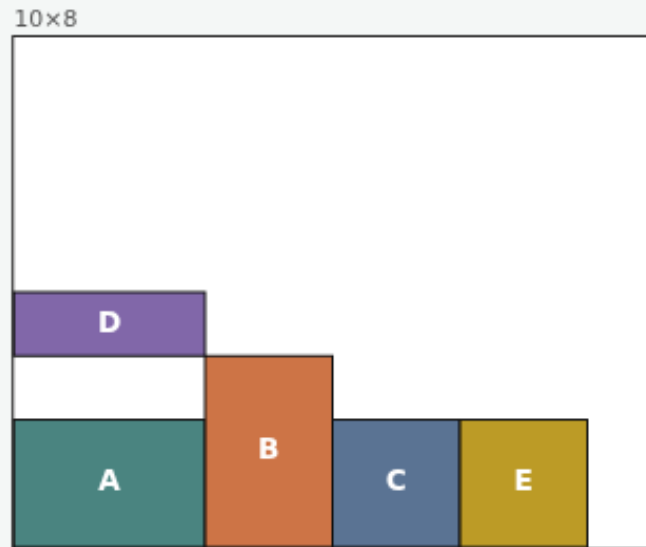
Golden SP packs legally

SP neighbors are permutation moves

SEQUENCE PAIR

SP neighbors are permutation moves

Step 5 / 5 · takeaway · module02-05-sequence-pair



Explanation

Annealing on sequence pair swaps or rotates entries in pos and neg. Feasibility stays geometric; cost can be deadspace plus wirelength.

- Neighbor = edit permutations
- Pack via longest paths
- Next: SA search on packings

starter golden: legal SP

SP neighbors are permutation moves

Browser lab track

- Open sequence-pair and Pack sequence pair
- Confirm five modules, non-negative coordinates

Implement track

- Implement longest-path SP packing
- Assert pos and neg are the same five ids, and the golden pair packs legally

Pitfalls

- Mismatched id sets between pos and neg

Your turn

- Pack the golden sequences legally
- Next: simulated annealing that prefers legal low-cost neighbors